Absorbance and Concentration · One complete topic

Absorbance and Concentration

Original instructional fixture in sources/lesson.md. No course or past-paper provenance is claimed. Generated 17 October 2026.

· How to use this pack

Read the definition and applicable conditions. Follow the worked solution, reproduce its check, and attempt the practice question before reading the answer. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Contents

1. 1 · Absorbance and Concentration
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

1 Absorbance and Concentration

Absorbance measures attenuation on a logarithmic scale. In the linear Beer–Lambert range it is proportional to concentration and path length. This lesson keeps fractions, percentages and physical units distinct so that the algebra matches the measurement. S1

Key terms and definitions

Absorbance. Negative base-ten logarithm of fractional transmittance. S1

Molar absorption coefficient. Wavelength- and species-dependent proportionality coefficient. S1

Formulas

E1.1 · Beer–Lambert relation

$$A = \varepsilon\, l\, c = -\log_{10} T$$

gives	Absorbance of a dilute absorbing species, such as illustrative Cu^{2+} in solution.
applies	Monochromatic measurement in the linear range with a suitable blank and consistent units.
units	A: dimensionless; ε: $\mathrm{L\,mol^{-1}\,cm^{-1}}$; l: cm; c: $\mathrm{mol\,L^{-1}}$.

S1

Graphs and diagrams

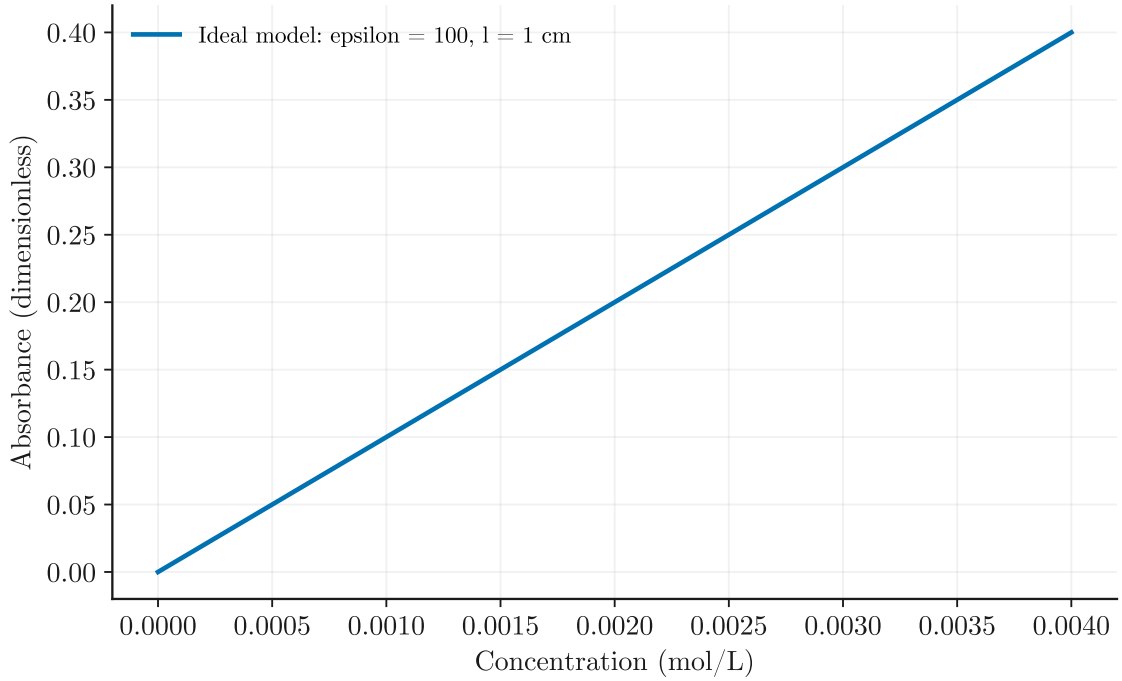


Figure G1.1. Absorbance increases linearly with concentration.

S1

How to read it. Read concentration on the horizontal axis and absorbance on the vertical axis. This is an ideal illustrative model in the stated linear range.

Flowcharts

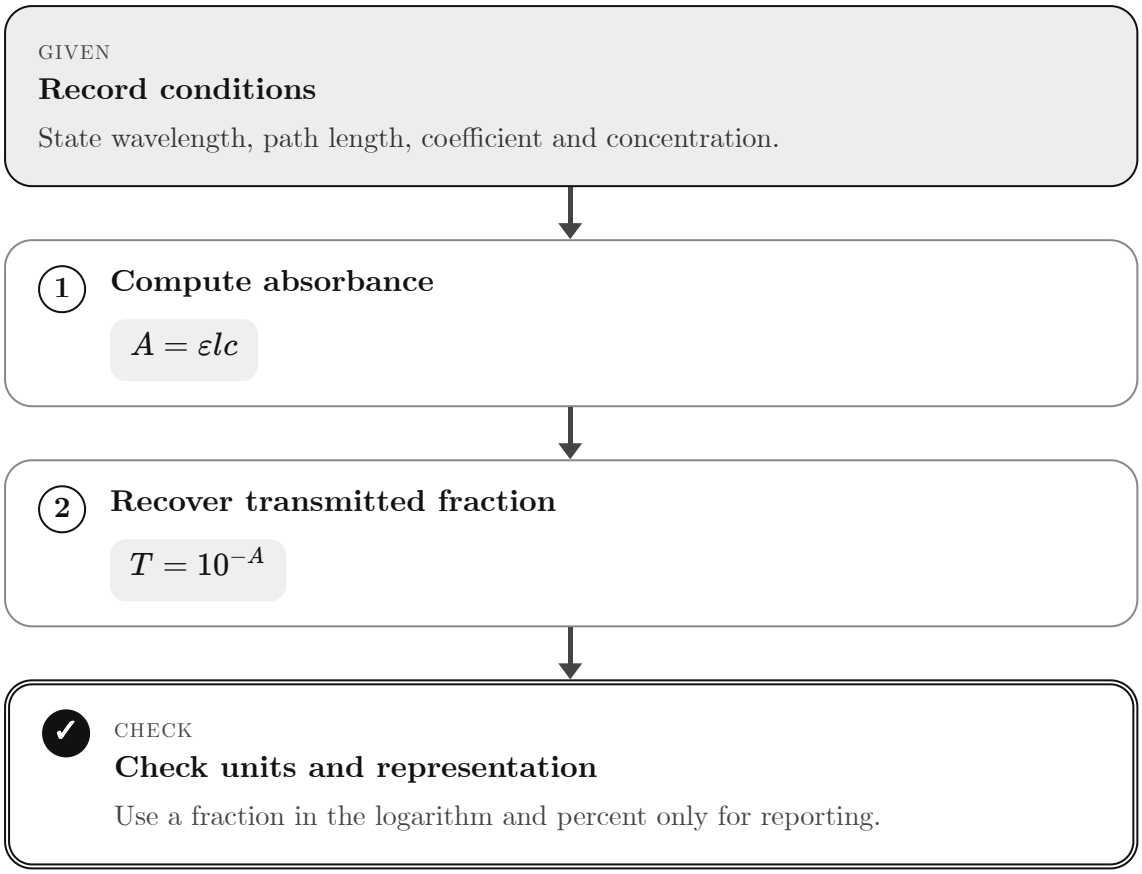


Figure F1.1. Apply and check the method

S1

Important points and why

1. Use the fractional transmittance in the logarithm. The illustrative coefficient is not a universal value for copper(II). S1

Why must fractional transmittance be used in the logarithm?
The definition uses I divided by I0. A percentage is one hundred times that ratio and would shift the logarithm by two.
S1

Worked examples

W1.1 · Convert absorbance to transmitted fraction

given	$\varepsilon = 100\ \mathrm{L\,mol^{-1}\,cm^{-1}}$, $l = 1\ \mathrm{cm}$, $c = 0.002\ \mathrm{mol\,L^{-1}}$.
method	Beer–Lambert calculation, then invert the base-ten logarithm.
step 1	$A = 100(1)(0.002) = 0.2$.
step 2	$T = 10^{-A} = 10^{-0.2} \approx 0.630957$.
step 3	Multiply the fraction by 100 to express a percentage.
answer	$A = 0.2$ and transmittance $\approx 63.0957\%$.
units	Absorbance and fractional transmittance are dimensionless.
check	Use the fractional transmittance in the logarithm. The illustrative coefficient is not a universal value for copper(II).

S1

Practice with explained answers

Q1.1 · Explain a condition before computing

Why must fractional transmittance be used in the logarithm?

S1

Answer · Q1.1

1. The definition uses I divided by I0. A percentage is one hundred times that ratio and would shift the logarithm by two.

Final answer. The definition uses I divided by I0. A percentage is one hundred times that ratio and would shift the logarithm by two.

Check. Check the answer against the explicit assumption in the included lesson.

short-note recap

Absorbance measures attenuation on a logarithmic scale. In the linear Beer–Lambert range it is proportional to concentration and path length. This lesson keeps fractions, percentages and physical units distinct so that the algebra matches the measurement.

S1

A Symbol glossary

Symbol	Name	Meaning	Unit	Appears in
A	Absorbance	Negative base-ten logarithm of fractional transmittance. S1	Dimensionless	E1.1
ε	Molar absorption coefficient	Proportionality coefficient at the specified species, wavelength and conditions. S1	$\mathrm{L\,mol^{-1}\,cm^{-1}}$	E1.1
l	Optical path length	Distance light travels through the sample. S1	cm	E1.1
c	Concentration	Amount of absorbing species per unit solution volume. S1	$\mathrm{mol\,L^{-1}}$	E1.1
T	Fractional transmittance	Ratio I/I_0; use the fraction, not its percentage. S1	Dimensionless	E1.1, W1.1
I, I_0	Transmitted and incident intensities	Both intensities use the same wavelength, geometry and units; their ratio is dimensionless. S1	Same intensity unit	E1.1

B Sources, coverage and gaps

Source	Path	Use
Original lesson	sources/lesson.md	All concepts and examples

No past papers supplied; practice is authored and unranked.

Known gaps, conflicts and corrections

Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

Evidence	Source unit	Status	Decision
S1	SRC1/lesson	covered	Read the entire original lesson and mapped definitions, assumptions, examples, limitations and output checks.

C Revision and retrieval practice

Keywords

Absorbance. Negative base-ten logarithm of fractional transmittance. S1

Molar absorption coefficient. Wavelength- and species-dependent proportionality coefficient. S1

Rapid recall

Why must fractional transmittance be used in the logarithm? The definition uses I divided by I0. A percentage is one hundred times that ratio and would shift the logarithm by two. S1

Essential facts

Use the fractional transmittance in the logarithm. The illustrative coefficient is not a universal value for copper(II). S1